



AWS Session

Summary - 19-3-2023

- If we want to use EKS we use eksctl tool:
 1. CLI
 2. Code (YAML)
- Nodegroup will manage the nodes. Nodegroup is managed by AWS. It will help in disaster recovery.
- ssm here is similar as ssh. Ssh-keygen creates public and private key.
- **YAML:** It is a generic way of formatting/providing information. It can be represented in the form of key-value pair. Equal spacing given in yaml file is called **indentation**.
- Nodegroup is one array which is in the form of hashes and dictionary.
- If we want to launch Kubernetes cluster using eksctl tool with the help of yml file then it means sending instruction which is called a declarative language. So yaml is a way to declare what we are declaring.

PRACTICAL

A screenshot of a Windows Command Prompt window titled "Command Prompt - ssh-keygen". The window shows the following text: "Microsoft Windows [Version 10.0.19044.2728]", "(c) Microsoft Corporation. All rights reserved.", and the command prompt "C:\Users\Vimal Daga>ssh-keygen". Below the command, it says "Generating public/private rsa key pair." and "Enter file in which to save the key (C:\Users\Vimal Daga\.ssh\id_rsa):". The cursor is positioned at the end of the prompt, ready for input. The background of the window is black, and the text is white.

[AWS]

```
Select Command Prompt
C:\Users\Vimal Daga/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Vimal Daga/.ssh/id_rsa.
Your public key has been saved in C:\Users\Vimal Daga/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:PSsfKqC1Hy0ERqAm1eJPzXVhEmZk7OUW6JZ/s3cG6mU vimal daga@DESKTOP-3E1AGGT
The key's randomart image is:
+---[RSA 3072]-----+
|  oo.  oBoo. |
| o... ++o+ |
| oo .oo + = . |
| o ....o =.o |
| o ..Soo |
| +. . .oo . |
| o oo o o. +E. |
| . . .o + .oo. o |
| .....o |
+---[SHA256]-----+
C:\Users\Vimal Daga>
```

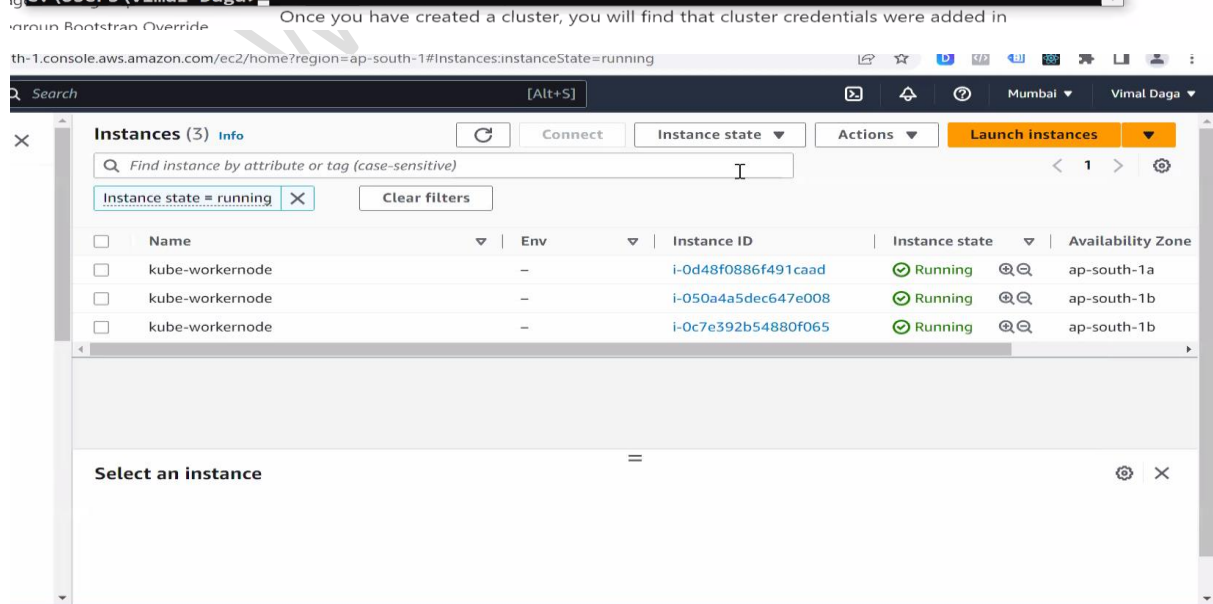
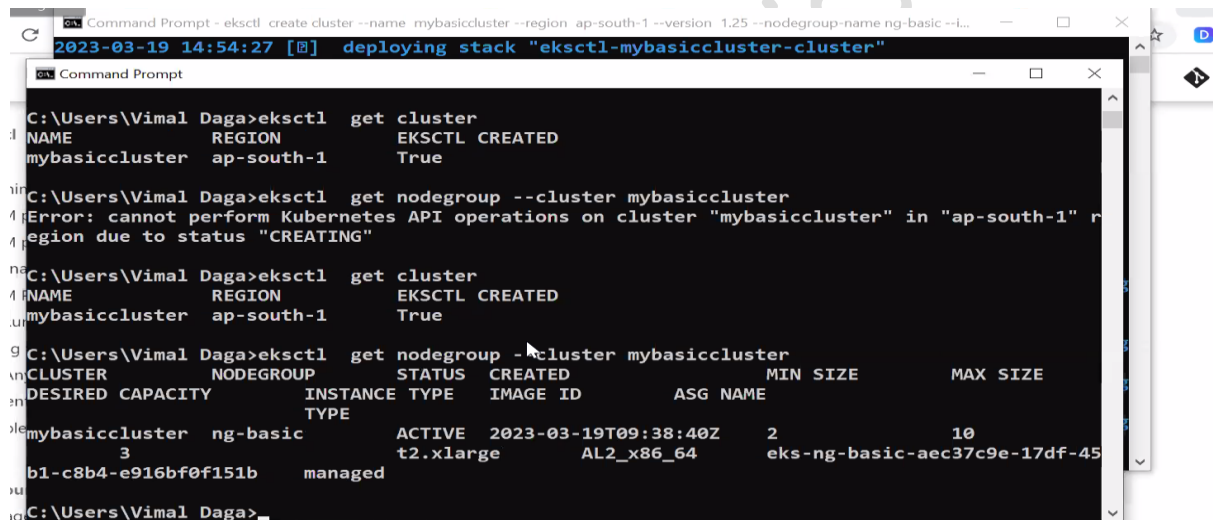
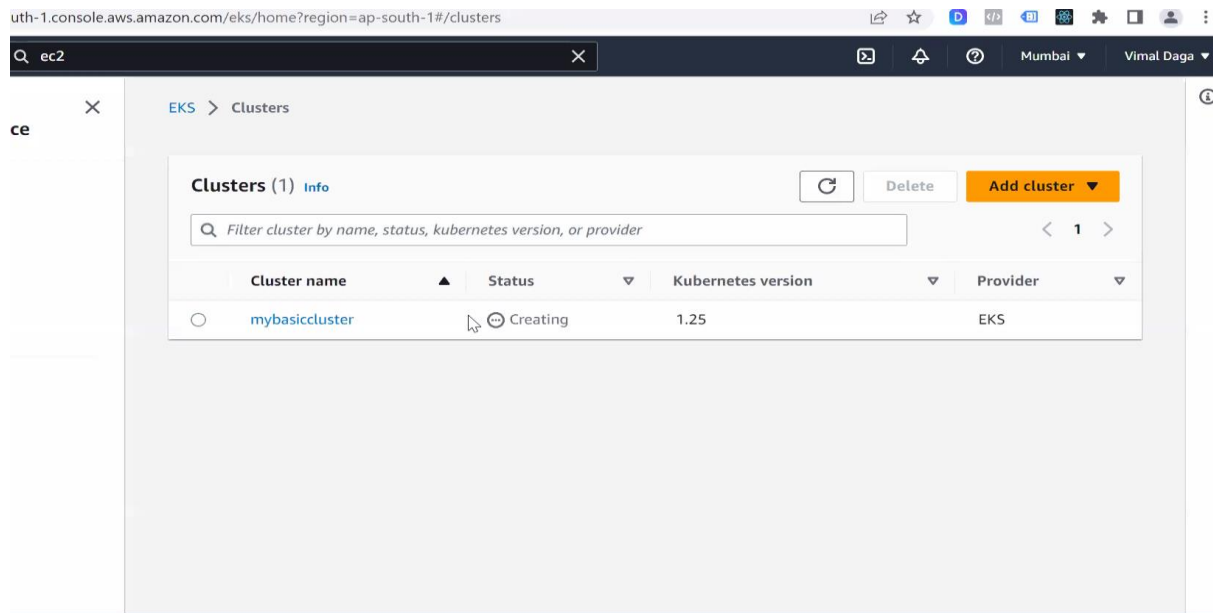
Now creating the cluster.

```
Command Prompt
Microsoft Windows [Version 10.0.19044.2728]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vimal Daga>eksctl create cluster --name mybasiccluster --region ap-south-1 --version 1.25 --nodegroup-name ng-basic --instance-types t2.xlarge --nodes 3 --nodes-min 2 --nodes-max 10 --node-volume-size 40 --node-volume-type gp3 --enable-ssm --ssh-access --instance-name kube-workernode --managed
```

```
es-max 10 --node-volume-size 40 --node-volume-type gp3 --enable-ssm --ssh-access --instance
-name kube-workernode --managed
2023-03-19 14:54:26 [!] SSM is now enabled by default; `ssh.enableSSM` is deprecated and will
1 be removed in a future release
2023-03-19 14:54:26 [i] eksctl version 0.133.0
2023-03-19 14:54:26 [i] using region ap-south-1
2023-03-19 14:54:26 [i] skipping ap-south-1c from selection because it doesn't support the f
ollowing instance type(s): t2.xlarge
2023-03-19 14:54:26 [i] setting availability zones to [ap-south-1a ap-south-1b]
2023-03-19 14:54:26 [i] subnets for ap-south-1a - public:192.168.0.0/19 private:192.168.64.0
/19
2023-03-19 14:54:26 [i] subnets for ap-south-1b - public:192.168.32.0/19 private:192.168.96.
0/19
2023-03-19 14:54:26 [i] nodegroup "ng-basic" will use "" [AmazonLinux2/1.25]
2023-03-19 14:54:26 [i] using SSH public key "C:\\Users\\Vimal Daga\\.ssh\\id_rsa.pub" as "eks
ctl-mybasiccluster-nodegroup-ng-basic-41:09:a3:7a:14:f9:38:c6:4a:25:c9:8a:d0:98:b2:ae"
2023-03-19 14:54:27 [i] using Kubernetes version 1.25
2023-03-19 14:54:27 [i] creating EKS cluster "mybasiccluster" in "ap-south-1" region with ma
naged nodes
2023-03-19 14:54:27 [i] will create 2 separate CloudFormation stacks for cluster itself and
the initial managed nodegroup
2023-03-19 14:54:27 [i] if you encounter any issues, check CloudFormation console or try 'ek
```

Cluster launched successfully.



← → ↻ ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#KeyPairs: [Alt+S]

aws Services Search [Alt+S] Mumbai

AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups

▼ Auto Scaling

Key pairs (1/2) Info

Search

	Name	Type	Created	Fingerprint
☐	mykey	rsa	2023/03/17 15:33 GMT+5:30	b6:bc:e3:1c:4b:85:c7:ae:fb
☒	eksctl-mybasiccluster-nodegroup-ng-ba...	rsa	2023/03/19 14:54 GMT+5:30	41:09:a3:7a:14:f9:38:c6:4a

Feedback Language © 2023, Amazon Web Services India Private Limited or its affiliates. Privacy Terms Co

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~
$
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~
$ ssh ec2-user@3.110.30.23
Last login: Sat Mar  4 23:45:20 2023 from 205.251.233.53

 _ _ | _ _ | _ _ |
 _ | ( _ | _ | /
 _ | \ _ | _ | _ |

Amazon Linux 2 AMI

https://aws.amazon.com/amazon-linux-2/
12 package(s) needed for security, out of 12 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-192-168-17-147 ~]$
[ec2-user@ip-192-168-17-147 ~]$
[ec2-user@ip-192-168-17-147 ~]$
[ec2-user@ip-192-168-17-147 ~]$ sudo su - root
[root@ip-192-168-17-147 ~]#
[root@ip-192-168-17-147 ~]# do
do      domainname done
[root@ip-192-168-17-147 ~]# do
do      domainname done
[root@ip-192-168-17-147 ~]# d

AMI Catalog      i-Od48f0886f491caad (kube-workernode)      3.110.30.23 | open address      192.168.17.147
                                                         Instance state      Public IPv4 DNS

Command Prompt: kubeconfig cluster-info dump
2023-03-19 15:11:47 [0] node "ip-192-168-52-254.ap-south-1.compute.internal" is ready
2023-03-19 15:11:49 [0] kubectl command should work with "C:\Users\Vimal Daga\.kube\config", try 'kubectl
get nodes'
2023-03-19 15:11:49 [0] EKS cluster "mybasiccluster" in "ap-south-1" region is ready

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl cluster-info
Kubernetes master is running at https://DDBA780968CA39F6D19535612EEA8C06.gr7.ap-south-1.eks.amazonaws.com
CoreDNS is running at https://DDBA780968CA39F6D19535612EEA8C06.gr7.ap-south-1.eks.amazonaws.com/api/v1/namespa
ces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

C:\Users\Vimal Daga>kubectl cluster-info dump
```

```

    },
    "items": []
  },
  {
    "kind": "PodList",
    "apiVersion": "v1",
    "metadata": {
      "resourceVersion": "4167"
    },
    "items": []
  }
]
}
}

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-17-147.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af
ip-192-168-36-145.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af
ip-192-168-52-254.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af

C:\Users\Vimal Daga>kubectl describe nodes ip-192-168-17-147.ap-south-1.compute.internal

```

avings Plans

```

Type      Reason              Age             From
Message
-----
-----
Normal    Starting            <unknown>
Warning   InvalidDiskCapacity 16m             kubelet, ip-192-168-17-147.ap-south-1.compute.internal
invalid capacity 0 on image filesystem
Normal    NodeHasSufficientMemory 16m (x2 over 16m) kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Node ip-192-168-17-147.ap-south-1.compute.internal status is now: NodeHasSufficientMemory
Normal    NodeHasNoDiskPressure 16m (x2 over 16m) kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Node ip-192-168-17-147.ap-south-1.compute.internal status is now: NodeHasNoDiskPressure
Normal    NodeHasSufficientPID 16m (x2 over 16m) kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Node ip-192-168-17-147.ap-south-1.compute.internal status is now: NodeHasSufficientPID
Normal    NodeAllocatableEnforced 16m             kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Updated Node Allocatable limit across pods
Normal    Starting            16m             kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Starting kubelet.
Normal    NodeReady           15m             kubelet, ip-192-168-17-147.ap-south-1.compute.internal
Node ip-192-168-17-147.ap-south-1.compute.internal status is now: NodeReady

C:\Users\Vimal Daga>

```

```

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-17-147.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af
ip-192-168-36-145.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af
ip-192-168-52-254.ap-south-1.compute.internal Ready    <none>   14m   v1.25.6-eks-48e63af

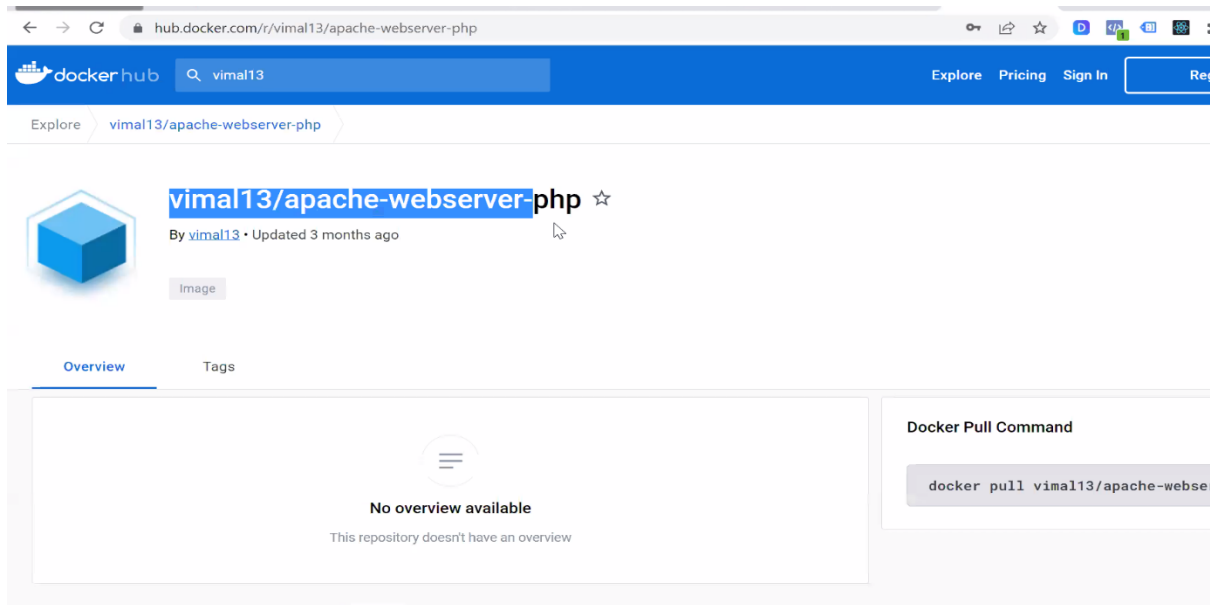
C:\Users\Vimal Daga>kubectl describe nodes ip-192-168-17-147.ap-south-1.compute.internal
Name:          ip-192-168-17-147.ap-south-1.compute.internal
Roles:         <none>
Labels:        alpha.eksctl.io/cluster-name=mybasiccluster
               alpha.eksctl.io/nodegroup-name=ng-basic
               beta.kubernetes.io/arch=amd64
               beta.kubernetes.io/instance-type=t2.xlarge
               beta.kubernetes.io/os=linux
               eks.amazonaws.com/capacityType=ON_DEMAND
               eks.amazonaws.com/nodegroup=ng-basic
               eks.amazonaws.com/nodegroup-image=ami-0287a728664c4cb7a
               eks.amazonaws.com/sourceLaunchTemplateId=lt-0190e4778ede65736
               eks.amazonaws.com/sourceLaunchTemplateVersion=1
               failure-domain.beta.kubernetes.io/region=ap-south-1
               failure-domain.beta.kubernetes.io/zone=ap-south-1a

```

avings Plans

[AWS]

- If we want to launch OS, server, app then that entire software we bundle in one box or software called **Image** and image in container world is called **container image**.
- We can create own images or if we want pre created image we can get from **hub.docker.com**



- If we want to launch app/container/pod with the help of image we call it as a **deployment** in k8s.
- Setup which k8s gives is highly resilient and fault tolerant.

```
C:\Users\Vimal Daga>kubectl get pods
No resources found in default namespace.

C:\Users\Vimal Daga>kubectl create deployment myweb --image=vimal13/apache-webserver-php
deployment.apps/myweb created

C:\Users\Vimal Daga>
```

7/21

```

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
myweb-764c9658b5-wqjv7             0/1     ContainerCreating   0           6s

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myweb-764c9658b5-wqjv7             1/1     Running   0           22s

C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS   ROLES    AGE   VERSION
ip-192-168-17-147.ap-south-1.compute.internal Ready   <none>   29m   v1.25.6-eks-48e63af
ip-192-168-36-145.ap-south-1.compute.internal Ready   <none>   29m   v1.25.6-eks-48e63af
ip-192-168-52-254.ap-south-1.compute.internal Ready   <none>   29m   v1.25.6-eks-48e63af

C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
myweb-764c9658b5-wqjv7             1/1     Running   0           2m1s  192.168.36.64  ip-192-168-52-254.ap-south-1.compute.internal

```

- Master keeps on monitoring the pod because there is a program running in worker nodes who communicates with the master, the program is known as **kubelet**. This is also managed by **EKS**.

```

C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
myweb-764c9658b5-wqjv7             1/1     Running   0           2m1s  192.168.36.64  ip-192-168-52-254.ap-south-1.compute.internal

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myweb-764c9658b5-wqjv7             1/1     Running   0           4m4s

C:\Users\Vimal Daga>kubectl delete pods myweb-764c9658b5-wqjv7
pod "myweb-764c9658b5-wqjv7" deleted

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
myweb-764c9658b5-hnm8f             0/1     ContainerCreating   0           4s

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
myweb-764c9658b5-hnm8f             0/1     ContainerCreating   0           7s

```

```

myweb-764c9658b5-hnm8f             1/1     Running   0           55s  192.168.33.121  ip-192-168-36-145.ap-south-1.compute.internal

```

```

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myweb-764c9658b5-hnm8f             1/1     Running   0           73s

C:\Users\Vimal Daga>kubectl describe pods myweb-764c9658b5-hnm8f
Name:          myweb-764c9658b5-hnm8f
Namespace:     default
Priority:       0
Node:          ip-192-168-36-145.ap-south-1.compute.internal/192.168.36.145
Start Time:    Sun, 19 Mar 2023 15:41:54 +0530
Labels:        app=myweb
               pod-template-hash=764c9658b5
Annotations:   <none>
Status:        Running
IP:            192.168.33.121
IPs:           IP: 192.168.33.121
Controlled By: ReplicaSet/myweb-764c9658b5

```

[AWS]

Now creating a deployment using image httpd

```
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl create deployment d1 --image=httpd
deployment.apps/d1 created

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
d1-67f7d5c54-f7ds6                 0/1     ContainerCreating   0           4s
myweb-764c9658b5-hnm8f             1/1     Running             0           6m14s

C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS             RESTARTS   AGE   IP              NODE
NOMINATED NODE   READINESS GATES
d1-67f7d5c54-f7ds6 1/1     Running            0           13s   192.168.36.64   ip-192-168-52-254.ap-south-1.compute.internal
<none>             <none>
myweb-764c9658b5-hnm8f 1/1     Running            0           6m23s   192.168.33.121 ip-192-168-36-145.ap-south-1.compute.internal
<none>             <none>

C:\Users\Vimal Daga>
```

```
C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS             RESTARTS   AGE   IP              NODE
NOMINATED NODE   READINESS GATES
d1-67f7d5c54-f7ds6 1/1     Running            0           13s   192.168.36.64   ip-192-168-52-254.ap-south-1.compute.internal
<none>             <none>
myweb-764c9658b5-hnm8f 1/1     Running            0           6m23s   192.168.33.121 ip-192-168-36-145.ap-south-1.compute.internal
<none>             <none>

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl exec -t -i myweb-764c9658b5-hnm8f bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl kubectl exec [POD] -- [COMMAND] instead.
[root@myweb-764c9658b5-hnm8f /]#
[root@myweb-764c9658b5-hnm8f /]#
[root@myweb-764c9658b5-hnm8f /]# ls
anaconda-post.log  boot  etc  lib  lost+found  mnt  proc  run  srv  tmp  var
bin                dev  home  lib64  media  opt  root  sbin  sys  usr
[root@myweb-764c9658b5-hnm8f /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 192.168.33.121 netmask 255.255.255.255 broadcast 0.0.0.0
    inet6 fe80::58a0:8aff:fe80:67e8 prefixlen 64 scopeid 0x20<link>
    ether 5a:a0:8a:80:67:e8 txqueuelen 0 (Ethernet)
```

Able to see the content.

```
[root@myweb-764c9658b5-hnm8f html]#
[root@myweb-764c9658b5-hnm8f html]#
[root@myweb-764c9658b5-hnm8f html]# curl http://192.168.36.64
<html><body><h1>It works!</h1></body></html>
[root@myweb-764c9658b5-hnm8f html]# exit
exit

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS             RESTARTS   AGE   IP              NODE
NOMINATED NODE   READINESS GATES
d1-67f7d5c54-f7ds6 1/1     Running            0           3m59s   192.168.36.64   ip-192-168-52-254.ap-south-1.compute.internal
<none>             <none>
myweb-764c9658b5-hnm8f 1/1     Running            0           10m   192.168.33.121 ip-192-168-36-145.ap-south-1.compute.internal
<none>             <none>

C:\Users\Vimal Daga>eksctl get cluster
NAME      REGION    EKSCTL CREATED
mybasiccluster ap-south-1 True

C:\Users\Vimal Daga>
```



```

Select Command Prompt
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
d1-67f7d5c54-f7ds6                 1/1     Running   0           28m
myweb-764c9658b5-hnm8f             1/1     Running   0           34m

C:\Users\Vimal Daga>kubectl get deployment
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
d1      1/1     1            1           32m
myweb   1/1     1            1           42m

C:\Users\Vimal Daga>kubectl scale deployment myweb --replicas=4
deployment.apps/myweb scaled

C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
d1-67f7d5c54-f7ds6                 1/1     Running             0           32m
myweb-764c9658b5-hnm8f             1/1     Running             0           39m
myweb-764c9658b5-kgw7g             0/1     ContainerCreating   0           7s
myweb-764c9658b5-psgnp             0/1     ContainerCreating   0           7s
myweb-764c9658b5-s5f6t             1/1     Running             0           7s

C:\Users\Vimal Daga>
myweb-764c9658b5-s5f6t             1/1     Running             0           18s

C:\Users\Vimal Daga>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
d1-67f7d5c54-f7ds6                 1/1     Running   0           33m   192.168.36.64   ip-192-168-52-254.ap-south-1.compute.internal
myweb-764c9658b5-hnm8f             1/1     Running   0           39m   192.168.33.121  ip-192-168-36-145.ap-south-1.compute.internal
myweb-764c9658b5-kgw7g             1/1     Running   0           37s   192.168.15.219  ip-192-168-17-147.ap-south-1.compute.internal
myweb-764c9658b5-psgnp             1/1     Running   0           37s   192.168.13.25   ip-192-168-17-147.ap-south-1.compute.internal
myweb-764c9658b5-s5f6t             1/1     Running   0           37s   192.168.52.103  ip-192-168-52-254.ap-south-1.compute.internal

C:\Users\Vimal Daga>kubectl get deployment
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
d1      1/1     1            1           34m
myweb   4/4     4            4           44m

C:\Users\Vimal Daga>

```

To check deployments use kubectl get deployment

```

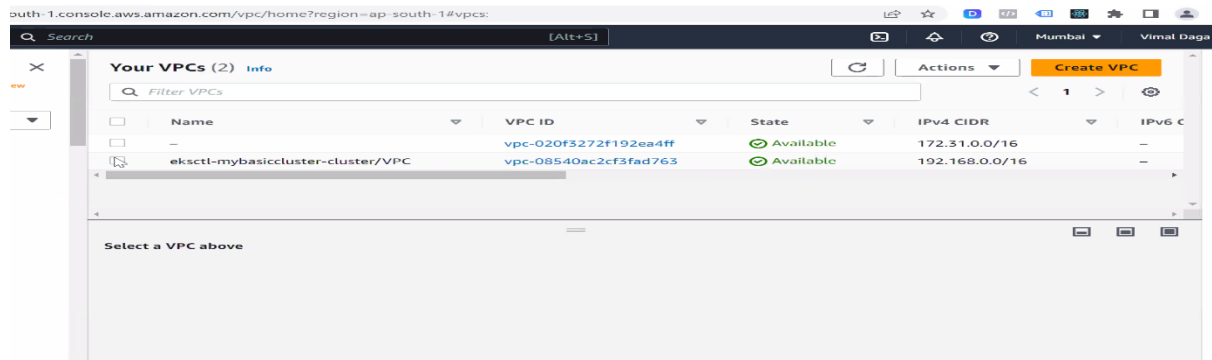
C:\Users\Vimal Daga>kubectl get deployment
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
d1      1/1     1            1           34m
myweb   4/4     4            4           44m

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>kubectl get svc
NAME            TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes      ClusterIP   10.100.0.1   <none>         443/TCP    83m

C:\Users\Vimal Daga>

```

- AWS has its own personal plugin called VPC Pugin for k8s cluster. So whenever we launch a cluster with EKS it launches VPC. Every VPC belongs to a subnet and every subnet gives a range of IP address. Every subnet has a personal range of network.



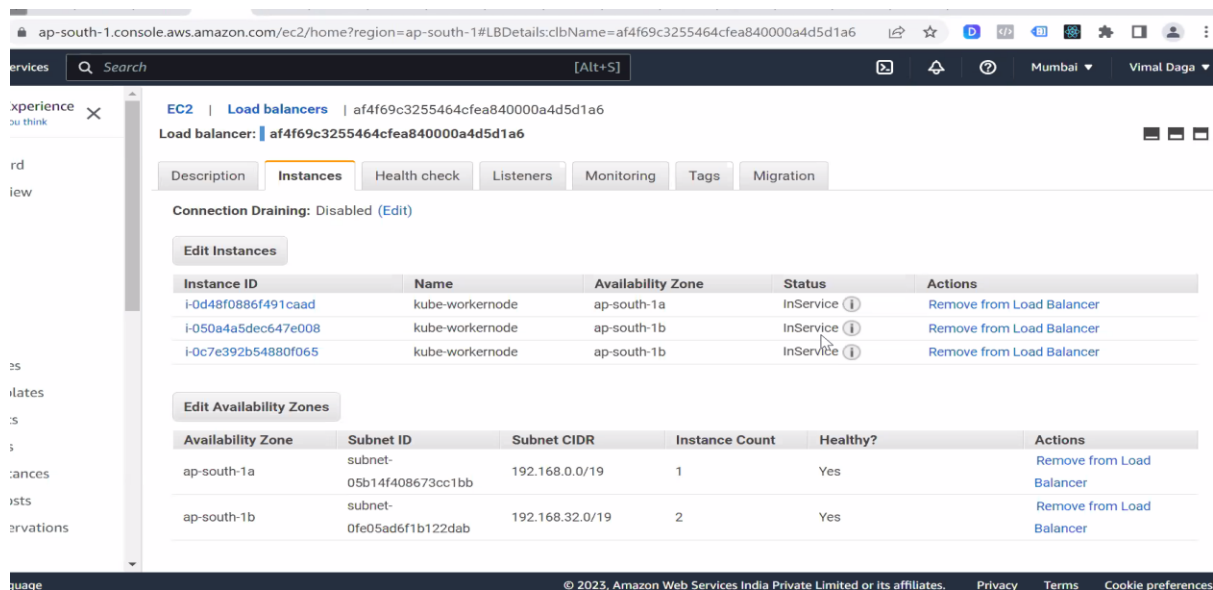
- AWS has a service called internet gateway which allows anybody entire that IP range to access the app. eksctl also creates the internet gateway.
- EKS has its own internal Load balancer. LB in Kubernetes is called a svc(service) that is created by a program which have some limit.
- In EKS load balancer is **Elastic load balancer-ELB**.
Other types of LB in AWS are:
 1. NLB- Network load balancer (gives high speed)
 2. ALB- Application load balancer (gives ingress setup)
 3. GLB- Gateway load balancer (Advanced setup that manages traffic, encrypt traffic, monitoring)
- For exposing a service we use `kubectl expose deployment myweb --name=mylbsvc --type=LoadBalancer --port 80`

Here **expose** means anybody can able to see the application over the internet.

```
C:\Users\Vimal Daga>kubectl expose deployment myweb --name=mylbsvc --type=LoadBalancer --port 80
service/mylbsvc exposed

C:\Users\Vimal Daga>kubectl get svc
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)
kubernetes    ClusterIP     10.100.0.1    <none>          443/TCP
mylbsvc       LoadBalancer  10.100.240.36 af4f69c3255464cfea840000a4d5d1a6-704834664.ap-south-1.elb.amazonaws.com 80:30401/TCP
```

[AWS]



ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LBDetails:clbName=af4f69c3255464cfea840000a4d5d1a6

services Search [Alt+S]

EC2 | Load balancers | af4f69c3255464cfea840000a4d5d1a6

Load balancer: af4f69c3255464cfea840000a4d5d1a6

Description Instances Health check Listeners Monitoring Tags Migration

Connection Draining: Disabled (Edit)

Edit Instances

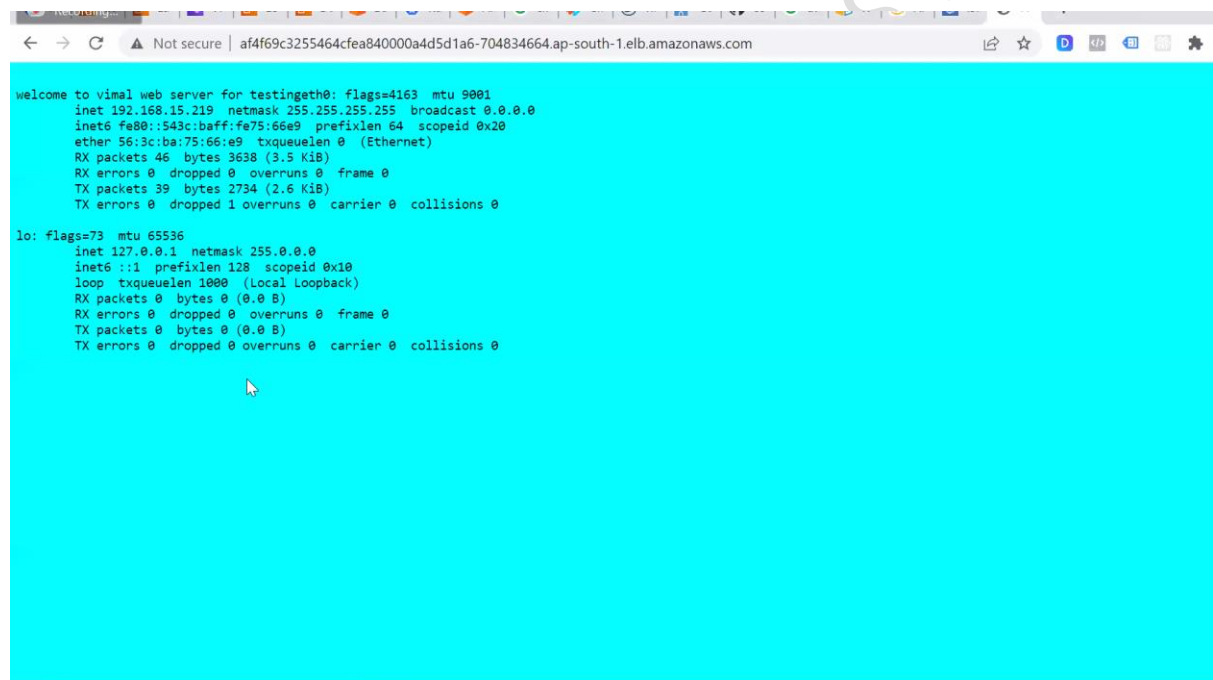
Instance ID	Name	Availability Zone	Status	Actions
i-0d48f0886f491caad	kube-workernode	ap-south-1a	InService ⓘ	Remove from Load Balancer
i-050a4a5dec647e008	kube-workernode	ap-south-1b	InService ⓘ	Remove from Load Balancer
i-0c7e392b54880f065	kube-workernode	ap-south-1b	InService ⓘ	Remove from Load Balancer

Edit Availability Zones

Availability Zone	Subnet ID	Subnet CIDR	Instance Count	Healthy?	Actions
ap-south-1a	subnet-05b14f408673cc1bb	192.168.0.0/19	1	Yes	Remove from Load Balancer
ap-south-1b	subnet-0fe05ad6f1b122dab	192.168.32.0/19	2	Yes	Remove from Load Balancer

© 2023, Amazon Web Services India Private Limited or its affiliates. Privacy Terms Cookie preferences

Now we can able to access the webpage successfully.



Not secure | af4f69c3255464cfea840000a4d5d1a6-704834664.ap-south-1.elb.amazonaws.com

```
welcome to vimal web server for testingeth0: flags=4163 mtu 9001
inet 192.168.15.219 netmask 255.255.255.255 broadcast 0.0.0.0
inet6 fe80::543c:baff:fe75:66e9 prefixlen 64 scopeid 0x20
ether 56:3c:ba:75:66:e9 txqueuelen 0 (Ethernet)
RX packets 46 bytes 3638 (3.5 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 39 bytes 2734 (2.6 KiB)
TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0

lo: flags=73 mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10
loop txqueuelen 1000 (Local Loopback)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



```
C:\Users\Vimal Daga>kubect1 get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP
kubernetes           ClusterIP   10.100.0.1     <none>
mylbsvc              LoadBalancer 10.100.240.36  af4f69c3255464cfea840000a4d5d1a6-704834664.ap-south-1.elb.amazonaws.com/TCP
113m

C:\Users\Vimal Daga>kubect1 get pvc
No resources found in default namespace.

C:\Users\Vimal Daga>kubect1 get sc
NAME                PROVISIONER      RECLAIMPOLICY   VOLUMEBINDINGMODE  ALLOWVOLUMEEXPANSION  AGE
vgp2 (default)      kubernetes.io/aws-efs  Delete          WaitForFirstConsumer  false                 115m

C:\Users\Vimal Daga>
```

[AWS]

```
C:\Users\Vimal Daga>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
d1-67f7d5c54-f7ds6                 1/1     Running   0           70m
myweb-764c9658b5-hnm8f             1/1     Running   0           76m
myweb-764c9658b5-kgw7g             1/1     Running   0           38m
myweb-764c9658b5-psgnp             1/1     Running   0           38m
myweb-764c9658b5-s5f6t             1/1     Running   0           38m

C:\Users\Vimal Daga>eksctl get cluster
NAME      REGION    EKSTL CREATED
mybasiccluster  ap-south-1  True

C:\Users\Vimal Daga>eksctl get nodegroup --cluster mybasiccluster
CLUSTER    NODEGROUP    STATUS  CREATED              MIN SIZE    MAX SIZE    DESIRED CAPACITY
CE TYPE IMAGE ID    ASG NAME
mybasiccluster  ng-basic     ACTIVE  2023-03-19T09:38:40Z  2           10          3
rge AL2_x86_64     eks-ng-basic-aec37c9e-17df-45b1-c8b4-e916bf0f151b  managed
```

```
C:\Users\Vimal Daga>eksctl scale nodegroup --cluster mybasiccluster --name ng-basic --nodes 4
2023-03-19 17:02:43 [0] scaling nodegroup "ng-basic" in cluster mybasiccluster
2023-03-19 17:02:44 [0] waiting for scaling of nodegroup "ng-basic" to complete
2023-03-19 17:04:07 [0] nodegroup successfully scaled

C:\Users\Vimal Daga>
C:\Users\Vimal Daga>
C:\Users\Vimal Daga>eksctl get nodegroup --cluster mybasiccluster
CLUSTER    NODEGROUP    STATUS  CREATED              MIN SIZE    MAX SIZE    DESIRED CAPACITY    INSTAN
CE TYPE IMAGE ID    ASG NAME
mybasiccluster  ng-basic     ACTIVE  2023-03-19T09:38:40Z  2           10          4                   t2.xla
rge AL2_x86_64     eks-ng-basic-aec37c9e-17df-45b1-c8b4-e916bf0f151b  managed

C:\Users\Vimal Daga>kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-17-147.ap-south-1.compute.internal  Ready    <none>   114m  v1.25.6-eks-48e63af
ip-192-168-31-38.ap-south-1.compute.internal   NotReady <none>   13s   v1.25.6-eks-48e63af
ip-192-168-36-145.ap-south-1.compute.internal  Ready    <none>   114m  v1.25.6-eks-48e63af
ip-192-168-52-254.ap-south-1.compute.internal  Ready    <none>   114m  v1.25.6-eks-48e63af

C:\Users\Vimal Daga>
```